

EXERCISE SET 3.1*

3.1.3 Let $R(m, n)$ be the predicate “If m is a factor of n then m is a factor of n ,” with domain for both m and n being $\mathbf{Z}$ the set of integers.

- a. Explain why $R(m, n)$ is false if $m = 25$ and $n = 10$.
- b. Give values different from those in part (a) for which $R(m, n)$ is false.
- c. Explain why $R(m, n)$ is true if $m = 5$ and $n = 10$.
- d. Give values different from those in (c) for which $R(m, n)$ is true.

“If m is a factor of n^2 , then m is a factor of n ”

$m, n \in \mathbf{Z}$

$R(m, n) = m \mid n^2 \rightarrow m \mid n$

- a. Explain why $R(m, n)$ is false if $m = 25$ and $n = 10$.

$$m = 25$$

$$n = 10$$

$$n^2 = 10^2 = 100$$

$$25 \mid 100 \rightarrow 25 \mid 10$$

$$100 / 25 = 4 \in \mathbf{Z}$$

$$10 / 25 = 2 / 5 = 0.4 \notin \mathbf{Z}$$

Solution: For $m = 25$ and $n = 10$, $R(m, n)$ is False, because $10^2 / 25 = 4 \in \mathbf{Z}$, whereas $10 / 25 = 0.4 \notin \mathbf{Z}$

b. Give values different from those in part (a) for which $R(m, n)$ is false.

$$m = 25$$

$$n = 5$$

$$n^2 = 5^2 = 25$$

$$25 \mid 25 \rightarrow 25 \mid 5$$

$$25 / 25 = 1 \in \mathbf{Z}$$

$$5 / 25 = 1 / 5 = 0.2 \notin \mathbf{Z}$$

Solution: For $m = 25$ and $n = 5$, $R(m, n)$ is False, because $5^2 / 25 = 1 \in \mathbf{Z}$, whereas $5 / 25 = 0.2 \notin \mathbf{Z}$

c. Explain why $R(m, n)$ is true if $m = 5$ and $n = 10$.

$$m = 5$$

$$n = 10$$

$$n^2 = 10^2 = 100$$

$$5 \mid 100 \rightarrow 5 \mid 10$$

$$100 / 5 = 20 \in \mathbf{Z}$$

$$10 / 5 = 2 \in \mathbf{Z}$$

Solution: For $m = 5$ and $n = 10$, $R(m, n)$ is True, because $10^2 / 5 = 20 \in \mathbf{Z}$, and $10 / 5 = 2 \in \mathbf{Z}$

d. Give values different from those in (c) for which $R(m, n)$ is true.

$$m = 3$$

$$n = 9$$

$$n^2 = 9^2 = 81$$

$$3 \mid 81 \rightarrow 3 \mid 9$$

$$81 / 3 = 27 \in \mathbf{Z}$$

$$9 / 3 = 3 \in \mathbf{Z}$$

Solution: For $m = 3$ and $n = 9$, $R(m, n)$ is True, because $9^2 / 3 = 27 \in \mathbf{Z}$, and $9 / 3 = 3 \in \mathbf{Z}$

Solutions: **a.** For $m = 25$ and $n = 10$, $R(m, n)$ is False, because $10^2 / 25 = 4 \in \mathbf{Z}$, whereas $10 / 25 = 0.4 \notin \mathbf{Z}$

b. For $m = 25$ and $n = 5$, $R(m, n)$ is False, because $5^2 / 25 = 1 \in \mathbf{Z}$, whereas $5 / 25 = 0.2 \notin \mathbf{Z}$

c. For $m = 5$ and $n = 10$, $R(m, n)$ is True, because $10^2 / 5 = 20 \in \mathbf{Z}$, and $10 / 5 = 2 \in \mathbf{Z}$

d. For $m = 3$ and $n = 9$, $R(m, n)$ is True, because $9^2 / 3 = 27 \in \mathbf{Z}$, and $9 / 3 = 3 \in \mathbf{Z}$